

OAL / ONE INDUSTRIES / LEWIS CORP

Accessibility Risk and Institutional Nonresponse Laboratory (ARIN Lab)

Independent Laboratory Charter, Experimental Framework, and Simulation Governance Specification

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Purpose: Establish an independent, executable laboratory charter for studying a fictional mandatory Android unlock animation, possible neurologic and accessibility harms, repeated exposure, population-scale rarity, nonrepresentative sampling, complaint undercounting, and persistent developer refusal to acknowledge or mitigate the issue.

What this document is: A proposed laboratory charter and simulation-governance framework derived from review of the attached multi-AI pass-through and the OAL / One Industries / Lewis Corp Documentation Governance Standard v0.1.

What this document is not: It is not a finding that any current Android animation causes seizures; not an accusation about Google, Android, or any real developer; not medical advice; not a clinical protocol for exposing people to flashing stimuli; not a legal conclusion; and not evidence that any simulated event actually occurred.

Executive Summary

The commission asks for more than a prevalence calculation or a one-off Monte Carlo model. It asks for a laboratory capable of testing a compound systems problem: a repeated visual exposure is

embedded in an unavoidable operating-system action; only a small susceptible subgroup may be at risk; the event can be rare enough to disappear inside an ordinary person's experience while becoming observable at platform scale; users, relatives, caregivers, and bystanders may all be exposed; and the fictional developer is stipulated to remain unwilling to acknowledge or mitigate the problem before, during, and after complaints, scientific evidence, injury, litigation, or adverse legal outcomes.

The most important design decision is to separate two propositions that must not be conflated. The **medical-causal proposition remains testable and falsifiable**: the laboratory must be allowed to conclude that the animation has no detectable seizure effect, that the effect exists only in a narrow susceptibility band, or that available evidence is insufficient. The **institutional-response proposition is a fixed scenario boundary**: if evidence of harm accumulates, the fictional organization still does not voluntarily implement meaningful mitigation. This prevents the simulation from prejudging medical causation while faithfully preserving the requested nonresponse scenario.

The ARIN Lab therefore operates as a dual-system laboratory. The **Harm and Evidence System (HES)** models exposure, susceptibility, adverse events, causal evidence, reporting, underreporting, and population outcomes. The **Institutional Nonresponse System (INS)** models denial, causal deflection, statistical minimization, complaint suppression, procedural resistance, symbolic compliance, litigation behavior, and post-judgment non-mitigation. HES can move from null evidence to strong evidence. INS does not automatically transition to corrective action. External forces may alter market behavior, regulation, litigation, user avoidance, or third-party controls, but voluntary developer mitigation remains disabled in the primary scenario.

The laboratory is explicitly simulation-first and non-harmful. It must not generate or display potentially provocative flashing content to human participants simply to validate the model. Where physical characteristics of a stimulus are needed, they should come from standards, instrumented non-human measurements, safe synthetic parameterization, or already-published data. Human health outcomes are modeled from sourced epidemiology and clearly labeled hypothetical assumptions rather than induced experimentally.

The laboratory's first mission is not to prove wrongdoing. Its first mission is to make the problem **measurable, falsifiable, reproducible, adversarially reviewable, and resistant to sampling illusions**. A worker with a small caseload can truthfully never witness a rare event while a global platform can still produce many events. Conversely, one observed event after an unlock does not establish causation. The framework is designed to preserve both truths at the same time.

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1. Commission and Laboratory Purpose

The ARIN Lab is commissioned to design and execute reproducible simulations around a fictional mobile operating-system behavior: a mandatory unlock animation containing a flash-like or abrupt luminance-transition effect that users cannot fully disable. The laboratory must be able to combine epidemiologic baselines, smartphone-use behavior, repeated-exposure mathematics, susceptibility distributions, adverse-event modeling, family and bystander exposure, support-service environments, reporting attrition, legal escalation, and long-term institutional refusal.

The laboratory exists because the problem cannot be answered by one average. The number of people with epilepsy does not by itself establish the number susceptible to a particular visual stimulus. The number of daily phone pickups does not automatically equal unlocks. A global user base does not resemble one person's social network. A complaint count does not equal an event count. A statistically rare event can be negligible inside a group of ten people and substantial across

billions of repeated exposures. The lab's job is to preserve these distinctions and quantify their interaction.

The intended outcome is a reusable research environment that can answer not only “how many events occur?” but also “what evidence would be visible, to whom, after how much exposure, under which assumptions, and what happens if the responsible organization refuses to learn from that evidence?”

2. Source Review and Independent Synthesis

The source package contains an original prompt, a detailed first-pass research design, a corrective prompt emphasizing persistent institutional refusal, and responses from multiple AI systems. The systems substantially converge on the central correction: the developer's refusal must be a **fixed condition of the fictional scenario**, not a temporary delay that disappears once a complaint threshold or lawsuit is reached.

The independent synthesis preserves the strongest distinct contributions without treating the set as automatically unanimous or canonical.

Table 2.1 - Contribution map from the reviewed AI responses

Source contribution	Retained laboratory value
ChatGPT	Separate the harm/evidence system from the organizational-response system; keep medical causation falsifiable while fixing nonresponse as a scenario premise; use counterfactual mitigation branches without making them the actual developer path.
Claude	Make the probability of voluntary fix/acknowledgment effectively zero at every escalation stage; track complaints filed versus answered, cumulative injury, legal actions, and months of continued nonresponse.
Perplexity	Formalize mitigation effectiveness as a parameter, including a primary nonresponse arm with mitigation effectiveness $M = 0$; model staged consequences from pre-release through post-litigation.
Grok	Treat internal observability as a separate failure mode: a system can create “absence of evidence” by failing to collect the signal or by recategorizing complaints; preserve the distinction between an asserted zero-risk arm and externally observed nonzero evidence.
Microsoft Copilot	Expand institutional behavior into explicit modes: denial, causal deflection, responsibility deflection, statistical minimization, technical minimization, accommodation refusal, complaint suppression, legal resistance, symbolic compliance, and post-judgment resistance.

The governing documentation standard adds a second layer: the work must identify simulation boundaries, distinguish sourced/observed/estimated/assumed/simulated values, retain provenance,

remain readable on mobile devices, preserve dissent, use durable item references, and not claim that a test or validation occurred when it did not.

3. Laboratory Charter

3.1 Name

Accessibility Risk and Institutional Nonresponse Laboratory (ARIN Lab).

The name intentionally describes the research function rather than presupposing a medical outcome. “Risk” permits a null result. “Institutional Nonresponse” identifies the fixed organizational scenario boundary.

3.2 Mission

To design, execute, audit, and preserve reproducible simulations that evaluate rare or subgroup-concentrated accessibility and health risks arising from repeated digital-interface exposure, with particular attention to how population-scale outcomes, personal observation, reporting systems, and institutional nonresponse can diverge.

3.3 Mandate

The lab may:

- establish historical and projected population baselines;
- model device-user populations and exposure frequency;
- model susceptible and nonsusceptible subgroups;
- simulate user, family, coworker, caregiver, and bystander exposures;
- test rare-event detectability at personal, organizational, and platform scales;
- model underreporting and complaint reclassification;
- simulate persistent refusal and external escalation;
- compare actual nonresponse with counterfactual mitigation controls;
- publish reproducible methods, assumptions, outputs, and failure cases.

The lab may not convert a hypothetical parameter into a factual medical rate merely because it was used in a simulation.

3.4 Operating principles

The ARIN Lab adopts the following principles:

1. **Causation is earned, not assumed.** Temporal association and simulated assignment are not proof of real-world causation.
2. **Nonresponse is a scenario boundary, not a medical conclusion.** Institutional refusal is fixed; biological effect remains testable.
3. **Rare does not mean nonexistent.** Platform-scale exposure can make very rare events observable.
4. **Observed absence is sample-dependent.** A small personal network can miss a real low-prevalence phenomenon.
5. **One event is not proof.** A witnessed event can justify investigation without establishing causation.

6. **No silent denominator changes.** Users, devices, unlocks, pickups, exposures, susceptible persons, events, reports, and confirmed cases remain distinct quantities.
7. **No invisible attrition.** Users who leave a device, reduce use, or stop reporting remain part of the causal and measurement story.
8. **Counterfactuals are controls, not fictional concessions.** Safer branches quantify preventability even though the primary developer does not adopt them.
9. **No human provocation requirement.** The lab does not need to expose people to hazardous flashing stimuli to validate a population simulation.
10. **Failure is preserved.** Null results, broken models, rejected assumptions, and dissent remain part of the record.

4. Foundational Scenario Boundaries

4.1 Fixed fictional product condition

The primary scenario contains an operating-system unlock animation that is mandatory for affected devices, includes a flash-like or rapid luminance-transition characteristic, and cannot be fully removed through ordinary accessibility, reduced-motion, developer, or personalization controls.

The exact visual hazard level is **not fixed**. It is a variable. Some simulation arms may place it below recognized concern thresholds; others may place a subset of devices, brightness states, panels, or users into higher-risk conditions.

4.2 Fixed institutional condition

For the primary scenario, meaningful voluntary mitigation is disabled throughout the observation period. The fictional developer does not voluntarily add an effective opt-out, redesign, safe default, rollback, warning system, meaningful incident pathway, or comprehensive population-level fix in response to accumulating evidence.

This remains true after:

- early complaints;
- clinician or caregiver reports;
- accessibility advocacy;
- independent research;
- internal concern;
- medically documented events;
- injuries;
- regulatory inquiry;
- litigation;
- settlement;
- adverse judgment;
- public controversy.

Legal or external forces may still create consequences, but the lab must not silently rewrite the scenario into “the company eventually does the right thing.”

4.3 Open medical condition

The following remain open questions rather than premises:

- whether the animation is biologically capable of triggering a seizure;
- whether only a photosensitive subgroup is affected;
- whether non-seizure neurologic or visual symptoms occur;
- whether repeated exposure changes individual risk;
- whether apparent event clustering exceeds background incidence;
- whether measured risk is zero, nonzero but tiny, or materially elevated.

4.4 Claim boundary

No result from this laboratory may be represented as a factual allegation about a real Android release or company unless a separate real-world evidence review is commissioned and completed.

5. Research Questions and Hypotheses

The laboratory organizes work around six primary research questions.

RQ1 - Baseline: How many people worldwide live with epilepsy or relevant seizure susceptibility, and how should historical, current, and projected figures be represented without conflating epilepsy, one-time seizures, photosensitivity, and visually induced events?

RQ2 - Exposure: How many unlock-animation exposures occur for light, typical, heavy, and very-heavy users, and how does repeated exposure scale from one person to the Android ecosystem?

RQ3 - Rare-event detectability: Under hypothetical per-exposure or per-susceptible-person risks, how many events would be expected, and what sample size is required to have a 50%, 90%, 95%, or 99% chance of observing at least one?

RQ4 - Sampling: How does encounter probability differ between the general population, low-contact networks, high-contact occupations, and nonrepresentative support-service environments?

RQ5 - Measurement failure: How do underreporting, complaint fragmentation, category mismatch, user abandonment, and missing telemetry cause internal dashboards to understate or completely miss real external events?

RQ6 - Persistent nonresponse: If evidence becomes increasingly strong but meaningful mitigation remains absent, how do preventable exposures, harms, legal/administrative consequences, user avoidance, inequity, and external intervention accumulate over time?

The lab must include null hypotheses. For example, a simulation arm may set animation-attributable seizure risk to zero while leaving background seizures present. This is essential for testing false attribution and for determining whether the analysis pipeline can distinguish background events from induced events.

6. Dual-System Architecture

The ARIN Lab uses two interacting but logically separate systems.

6.1 Harm and Evidence System (HES)

HES represents the outside world. It contains:

population -> Android adoption -> applicable devices -> unlock frequency -> animation exposures -> susceptible subgroup -> trigger-compatible exposures -> adverse responses -> medical outcomes -> observations -> reports -> evidence accumulation.

HES may produce zero or nonzero attributable events depending on the selected parameter arm. It also contains background seizure events so that the lab must solve a causal-attribution problem rather than merely count temporally adjacent events.

6.2 Institutional Nonresponse System (INS)

INS represents the fictional developer's information and decision system. It receives only a filtered subset of HES outputs. The filtration can include nonreporting, support-ticket misclassification, data fragmentation, requests for impossible proof, category definitions that exclude health events, and analytic choices that favor an asserted null.

The primary INS state is **NO MATERIAL CORRECTIVE ACTION**. Evidence can alter litigation cost, media attention, user trust, regulator attention, staff concern, market exit, or external restrictions without automatically altering this state.

6.3 Why the separation matters

A common modeling error would be to let strong evidence automatically force remediation. That would erase the very condition the commission wants studied. The opposite error would be to let institutional refusal prove that the hazard is real. The dual-system architecture prevents both errors.

7. Population and Exposure Model

The Population and Exposure Workstream builds the denominators required by every later calculation.

The model must distinguish at least:

- world population;
- people living with epilepsy;
- people with photosensitive epilepsy or relevant visual susceptibility;
- smartphone users;
- Android users;
- active Android devices;
- devices running the fictional affected software;
- users who actually see the animation;
- bystanders who see it;
- unlocks;
- phone pickups;
- screen activations;
- sessions;
- animation exposures.

Unlock frequency should be modeled as a distribution, not a single mean. At minimum, the lab should maintain light, typical, heavy, and extreme-use strata and permit age, occupation, notification load, assistive use, and work dependence to alter the distribution.

Repeated exposure must be calculated at daily, weekly, monthly, annual, and five-year scales. Five years is the initial primary horizon, but the architecture must permit longer runs because the fixed nonresponse scenario may persist beyond that point.

8. Susceptibility and Outcome Model

The lab uses layered susceptibility rather than assigning equal risk to every user.

A conceptual chain is:

Exposure -> susceptible person -> stimulus-compatible condition -> biologic response -> observable symptom -> seizure or non-seizure outcome -> secondary consequence.

The susceptibility model may include age, known epilepsy, photosensitivity, prior photoparoxysmal response, medication status, sleep deprivation, viewing distance, brightness, ambient light, panel behavior, animation timing, and other variables only when a source or explicit hypothetical assumption supports their inclusion.

Outcomes are separated into categories such as no response, visual discomfort, headache or migraine, dizziness/disorientation, aura-like symptoms, seizure without injury, seizure with medical evaluation, seizure with secondary injury, emergency treatment, hospitalization, and rare severe secondary outcomes.

The lab must never assign clinical frequencies to those outcomes without labeling the source category. A number may be **Published Estimate, Derived Calculation, Projection, Hypothetical Assumption, or Simulation Result**. These labels should travel with the variable into exported tables.

9. Representative and Nonrepresentative Sampling

The commission specifically requires the lab to study the difference between the world and the people one person actually encounters.

The Personal Encounter Module therefore models:

- low-contact personal networks;
- general-population contact networks;
- high-contact occupations;
- family/household exposure;
- caregiver and disability-service environments;
- other deliberately nonrepresentative populations.

The module must not assume that disability itself implies seizure susceptibility. Instead, it can run sourced or hypothetical prevalence multipliers and show how encounter probability changes. A sensitivity series might compare 1x, 2x, 3x, and 5x a general-population baseline, with each multiplier clearly labeled as an experimental assumption unless external evidence supports it.

The module must report both:

- probability that at least one susceptible person exists in the observed network; and

- probability that at least one animation-attributable event occurs during the observation period.

Those are not the same probability.

10. Institutional Nonresponse Model

Institutional refusal is represented as a state machine whose behavior changes form without changing its core outcome.

Table 10.1 - Primary nonresponse states

State	Behavior	Modeling consequence
N0 - Pre-release dismissal	Concerns not sought, not escalated, or treated as outside scope	Hazard, if present, reaches deployment
N1 - Anecdotalization	Individual reports treated as isolated or “not reproducible”	Event count and complaint count diverge
N2 - Causal deflection	Events acknowledged but attributed to background conditions	No product mitigation
N3 - Statistical minimization	Affected subgroup characterized as too small	Population denominator used to dilute subgroup risk
N4 - Technical minimization	Internal test or threshold language used as dispositive	Uncertainty narrowed selectively
N5 - Complaint suppression	Health complaints fragmented, recoded, or closed	Internal detection probability falls
N6 - Procedural/legal resistance	Claims delayed, contested, appealed, or settled without systemic change	External cost rises; exposure persists
N7 - Symbolic compliance	Guidance, statements, or narrow accommodations occur without removing exposure	Appearance of response without effective mitigation
N8 - Post-judgment persistence	Even after adverse action, underlying default remains	Preventable exposure continues after legal salience

The main scenario sets the **Mitigation Effectiveness Parameter $M = 0$** for voluntary developer action. Counterfactual control arms may set M between 0 and 1, but those controls are analytical comparisons rather than actions the fictional developer takes.

11. Evidence, Reporting, and Detection Model

The laboratory must model the path from a real event to an organizationally visible signal.

A useful decomposition is:

True event -> noticed -> attributed as possibly related -> documented -> reported -> accepted by intake -> correctly categorized -> aggregated -> analyzed -> escalated -> acted upon.

Every arrow can have a probability below 1. A system can therefore have many true events and few visible cases.

Required underreporting scenarios include 100%, 50%, 25%, 10%, 5%, and 1% reporting capture, plus compound-loss scenarios in which later stages reduce visibility further.

The model should explicitly test **measurement-created absence**: if telemetry records unlock counts but no symptom data, the absence of a symptom field cannot legitimately be treated as evidence that symptoms did not occur. Likewise, if a support system labels “flash made me sick” as a preference complaint, the dashboard can remain empty while the external event burden is nonzero.

12. Counterfactual and Control Arms

The primary branch is persistent nonresponse. It must be compared against controls so preventability can be measured.

Required arms:

- **C0 - Null causal arm:** attributable seizure risk = 0; background events remain.
- **C1 - Persistent nonresponse:** $M = 0$ for the full run.
- **C2 - Complete opt-out after first credible signal:** susceptible users can avoid the animation.
- **C3 - Reduced-motion integration:** exposure is reduced for users selecting an accessibility setting.
- **C4 - Safe-by-default redesign:** the potentially provocative characteristic is removed for all users.
- **C5 - Immediate recall/rollback:** affected version is removed rapidly.
- **C6 - External-only intervention:** developer remains resistant but an OEM, regulator, court, third party, or market mechanism reduces exposure.

The purpose of C2-C6 is not to soften C1. It is to quantify the difference between what happens and what could have been prevented.

13. Simulation Engine and Parameter Registry

The initial implementation should support deterministic calculations and Monte Carlo simulation.

Each synthetic person may carry variables for age band, device use, unlock-rate distribution, direct exposure, bystander exposure, epilepsy status, photosensitivity status, susceptibility state, triggering threshold, background seizure rate, animation-attributable event probability, outcome severity, reporting probability, and dropout/device-abandonment behavior.

Each simulation run must record:

- random seed;
- model version;
- parameter-set version;
- source category for every major input;
- scenario arm;
- population size or scaling method;
- execution timestamp;
- output file hashes where practical.

The engine should permit both individual-level simulation and analytic rare-event calculation. Extremely rare events can be computationally inefficient to simulate naively across billions of users; Poisson/binomial approximations or weighted simulation may be more appropriate. The method must be documented rather than hidden behind a single output number.

14. Experimental Exercise Program

The laboratory becomes credible by running exercises, not merely accumulating frameworks. The minimum program is defined in Appendix B. The core sequence is:

1. baseline population reconstruction;
2. unlock-exposure reconstruction;
3. rare-event detectability test;
4. representative-versus-nonrepresentative encounter test;
5. underreporting and complaint-visibility test;
6. five-year persistent-nonresponse run;
7. counterfactual mitigation comparison;
8. adversarial falsification run.

Each exercise requires a hypothesis/objective, setup, parameter table, procedure, execution record, outputs, anomalies, interpretation, and repeatability note. A proposed exercise must never be represented as completed until it has actually been executed.

15. Roles, Responsibilities, and Decision Rights

The lab separates execution from review.

Laboratory Director / Human Coordinator: defines commission scope, decides whether outputs become project-governing material, and resolves governance questions that cannot be delegated.

Primary Simulation Architect: maintains model architecture, scenario boundaries, parameter interfaces, and reproducibility requirements.

Epidemiology and Evidence Analyst: manages sourced prevalence/incidence inputs and distinguishes epilepsy, seizures, photosensitivity, and visually induced events.

Exposure and Device-Use Analyst: manages Android population estimates, unlock/pickup/session definitions, exposure distributions, and bystander exposure.

Statistical Modeler: implements rare-event mathematics, Monte Carlo methods, confidence intervals, sample-size analysis, and sensitivity analysis.

Accessibility and Human-Factors Reviewer: assesses whether assumptions and outputs reflect accessibility concerns without treating disability as a uniform risk category.

Institutional-Behavior Modeler: maintains the INS state machine, complaint attrition, nonresponse modes, and counterfactual external interventions.

Legal/Regulatory Scenario Reviewer: models procedural pathways and external consequences as fictional scenario mechanics; does not convert them into legal advice or findings about real entities.

Adversarial Reviewer: attempts to break the causal model, denominator choices, parameter assumptions, and organizational logic; preserves dissent.

Documentation and Reproducibility Steward: enforces identifiers, versioning, claim labels, cross-format fidelity, experiment logs, and archival completeness.

No role may certify its own high-stakes conclusion as independently validated. The modeler can execute; validation requires a separate review function.

16. Ethics, Safety, and Accessibility Rules

The ARIN Lab is a simulation and modeling environment, not a human provocation laboratory.

The following are prohibited as default validation methods:

- intentionally exposing known susceptible participants to flashing stimuli;
- generating hazardous flashing sequences for demonstration when parameterized modeling is sufficient;
- treating a simulated event as medical advice;
- inferring that an individual person's seizure was caused by a phone animation without clinical evidence;
- inferring that a disability-service population has elevated seizure prevalence merely because it is disabled;
- using fear-producing visuals as presentation decoration.

The laboratory should use non-harmful data sources and computational models wherever possible. Any future human-subject research would require separate ethical review, informed consent, clinical safety controls, and a protocol outside this charter.

17. Validation, Verification, and Falsification

Verification asks whether the code correctly implements the specified model. Validation asks whether the model is adequate for the research question. They are not interchangeable.

Minimum verification checks include unit tests for exposure arithmetic, probability bounds, seed reproducibility, denominator identity, reporting attrition, and counterfactual branch logic.

Minimum validation checks include comparison with published epidemiologic ranges, published device-use behavior where available, sensitivity to plausible parameter changes, and inspection for impossible outputs.

Falsification requirements are especially important. The model must be capable of producing and preserving at least four broad conclusions:

1. no detectable attributable effect;
2. effect too small to distinguish from background under available data;
3. subgroup-concentrated effect that is detectable only with targeted or platform-scale analysis;
4. material effect with substantial preventable harm.

If the engine can only produce the conclusion the researcher expects, it has failed as a laboratory instrument.

18. Adversarial Review and Failure Preservation

Every major run must receive an adversarial pass that asks:

- Did we mistake phone pickups for unlocks?
- Did we use a population-wide denominator to conceal subgroup risk?
- Did we overstate photosensitivity prevalence?
- Did we assume every flash-like event can trigger a seizure?
- Did we double-count repeated events in the same person?
- Did we ignore background seizure incidence?
- Did we assume exposures are independent when they are not?
- Did user dropout create survivorship bias?
- Did complaint filtering create artificial absence of evidence?
- Did we confuse temporal association with causation?
- Did the fixed nonresponse premise leak into the medical causal result?

Rejected models and failed runs should be archived with reasons rather than deleted. The laboratory should preserve disagreement between AI reviewers when the disagreement is substantive.

19. Outputs, Metrics, and Publication Package

Every full study should produce, at minimum:

- executive research summary;
- input and provenance register;
- historical/current/projected population table;
- unlock/exposure distribution;
- susceptibility assumptions;
- rare-event expected-value and sample-size calculations;
- representative/nonrepresentative encounter analysis;
- underreporting analysis;
- five-year persistent-nonresponse simulation;
- counterfactual preventability comparison;
- uncertainty and sensitivity analysis;
- limitations;
- adversarial review;
- reproducibility package;
- PDF, DOCX, and Markdown narrative outputs.

Core metrics include total exposures, susceptible exposures, attributable events, background events, repeat events, injuries, medical utilization, reports filed, reports accepted, correctly categorized reports, internal visible cases, time to first external signal, time from signal to legal escalation, unresolved complaints, user abandonment, cumulative preventable exposure, and counterfactual events avoided.

All charts, tables, and figures must be numbered and referenced by item number. The narrative must explain their meaning; no matrix or dashboard can substitute for interpretation.

20. Governance, Versioning, and Change Control

This charter is v0.1 and PROVISIONAL. It does not become canon or an adopted OAL standard merely because it exists.

Material changes require a revision note describing what changed and why. Changes to the fixed scenario boundaries, causal structure, or required experiment suite require at least a minor-version increment and adversarial review. A change that converts the primary scenario from persistent nonresponse into eventual voluntary remediation is a material scope change, not an editorial edit.

Parameter updates should be versioned separately from the charter when possible so that a new epidemiologic estimate does not force a rewrite of the laboratory constitution.

21. Limitations and Unresolved Questions

This charter does not yet supply the current global epilepsy dataset, photosensitivity estimates, Android active-user count, unlock-frequency evidence, or real-world legal analysis. Those belong to execution phases that require external research and source verification.

The largest unresolved scientific issue is that “flash-like” is not a sufficient technical stimulus description. Real risk depends on frequency, luminance change, contrast, field of view, viewing conditions, display technology, user susceptibility, and other parameters. A future technical annex should define a safe, non-provocative representation of those characteristics without requiring human exposure.

The largest unresolved systems issue is how to model external intervention when developer voluntary mitigation remains zero. Courts, regulators, OEMs, insurers, employers, app ecosystems, and user migration can all alter exposure without changing the developer's internal stance. Those mechanisms should be separate modules rather than hidden inside M.

22. Success Criteria and Next Actions

The ARIN Lab succeeds when a new reviewer can reproduce a run, identify every major assumption, distinguish fact from simulation, understand why personal observation differs from platform prevalence, inspect how reports disappear between event and dashboard, and compare persistent nonresponse with safer counterfactuals without confusing the two.

The immediate next actions are:

1. create a sourced epidemiology baseline for the previous five years, 2026, and five-year projections;
2. establish defensible Android user/device and phone-unlock distributions;
3. define the first parameter registry with source labels;
4. implement the null, persistent-nonresponse, and safe-default arms;
5. run the minimum exercise suite in Appendix B;
6. subject the results to independent adversarial review;
7. publish the run as a separate Experimental Exercise Report rather than claiming this charter itself is an executed simulation.

Appendix A - Core Variable Registry

Table A.1 - Minimum core variables

Variable	Meaning	Default source class
P_world	World population by year	Published estimate / projection
P_epi	People living with epilepsy	Published estimate / derived
P_photo	Photosensitive or visually susceptible subgroup	Published estimate / bounded assumption
P_android	Active Android users/devices	Published estimate / bounded estimate
U_day	Unlocks per user per day	Published behavior / modeled distribution
E_direct	Direct animation exposures	Derived calculation
E_bystander	Secondary/bystander exposures	Hypothetical / modeled
S_i	Individual susceptibility state	Synthetic simulation variable
R_bg	Background seizure-event rate	Published estimate / modeled
R_attr	Animation-attributable event rate	Hypothetical assumption unless evidence exists
Q_report	Probability an event is reported	Hypothetical/sourced sensitivity parameter
Q_classify	Probability a report is correctly categorized	Hypothetical sensitivity parameter
M	Voluntary mitigation effectiveness	Fixed at 0 in primary scenario
X_external	External intervention effectiveness	Scenario parameter
D_dropout	Device abandonment/use-reduction probability after adverse experience	Hypothetical / estimated

Appendix B - Minimum Exercise Suite

Exercise B1 - Baseline Reconstruction

Build the five-year historical, 2026 baseline, and five-year projected seizure/epilepsy dataset. Output must distinguish direct source values from interpolations and projections.

Exercise B2 - Exposure Ladder

Calculate cumulative animation exposure for light, typical, heavy, and extreme users and scale to the affected Android population. Include direct and bystander exposure separately.

Exercise B3 - Rare-Event Detection

For event assumptions ranging from extremely rare to deliberately high stress-test values, compute expected events and the sample/exposure count needed for 50%, 90%, 95%, and 99% probability of seeing at least one event.

Exercise B4 - Personal Network Test

Compare a small personal/caseload network with general-population and high-contact networks. Demonstrate how a true population event can remain unseen locally, and how one local event still fails to prove causation.

Exercise B5 - Complaint Visibility Collapse

Start with a known simulated event count and pass it through reporting, intake, categorization, aggregation, and escalation filters. Measure how many events remain visible internally at 100%, 50%, 25%, 10%, 5%, and 1% capture levels.

Exercise B6 - Five-Year Persistent Nonresponse

Run HES and INS together for five years with $M = 0$. Record evidence accumulation, external consequences, user adaptation, and cumulative preventable exposure without allowing automatic developer remediation.

Exercise B7 - Counterfactual Preventability

Replay the same random seed under C2-C6 controls. The difference in outcomes is the modeled preventable burden, not evidence that a real developer would have behaved that way.

Exercise B8 - Adversarial Falsification

Set attributable risk to zero, introduce background seizures and biased reporting, and test whether the pipeline falsely “discovers” causation. Then introduce a small subgroup effect and test whether broad population averaging hides it.

Appendix C - Publication and Acceptance Checklist

A full ARIN publication is not ready unless:

- purpose and intended outcome are explicit;
- status and simulation nature are visible;
- preparer and contributors are identified;
- fixed assumptions and open hypotheses are separated;
- sourced, calculated, projected, hypothetical, and simulated values are labeled;
- no human hazardous-exposure test is implied or required;
- tables and figures are numbered and interpreted;
- mobile readability is preserved;
- PDF, DOCX, and Markdown carry substantively equivalent content;
- the run identifier, seed, parameter version, and execution status are stated;
- failed or dissenting analyses are preserved;
- the document says what has actually been executed and what remains proposed;

- conclusions do not exceed the evidence.

Appendix D - Independent Review Record

This charter was independently prepared by ChatGPT - GPT-5.6 Sol after reviewing the user-provided multi-AI pass-through containing the original commission, corrective prompt, and responses from ChatGPT, Claude, Perplexity, Grok, and Microsoft Copilot, together with the attached OAL / One Industries / Lewis Corp Documentation Governance Standard v0.1.

The charter does not mechanically merge those responses. It adopts the common fixed-nonresponse premise, preserves the most useful distinct modeling contributions, and adds an independent laboratory architecture centered on falsifiability, dual-system separation, experiment execution, counterfactual controls, underreporting, and failure preservation.

Disposition: PROPOSED / PROVISIONAL. No simulation has been executed by this document. The next artifact should be an Experimental Exercise Report containing sourced inputs and actual run outputs.